2 (a) (Thermal) energy / heat required to convert unit mass of solid to liquid — M1
at its normal melting point / without any change in temperature — A1 — [2]
(*reference to 1 kg or to ice $\rightarrow$ water scores max 1 mark*)

(b) (i) To make allowance for heat gains from the atmosphere — B1 — [1]

(ii) e.g. constant rate of production of droplets from funnel
constant mass of water collected per minute in beaker
(*any sensible suggestion, 1 mark*) — B1 — [1]

(iii) mass melted by heater in 5 minutes = 64.7 – ½ × 16.6 = 56.4 g — C1
$56.4 \times 10^{-3} \times L = 18$ — C1
$L = 320\ \text{kJ kg}^{-1}$ — A1 — [3]
(*Use of m = 64.7, giving L = 278 kJ kg*$^{-1}$*, scores max 1 mark*
use of m = 48.1, giving L = 374 kJ kg^{-1}*, scores max 2 marks*)

3 (a) acceleration / force (directly) proportional to displacement — M1